Design Proposal: Mach-Zehnder Interferometer

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Abstract— **This project presents the design of a Mach–Zehnder interferometer (MZI) for integrated photonics applications. Using SiEPIC tools, Lumerical MODE, and Lumerical INTERCONNECT, the interferometer is optimized to achieve controlled output splitting ratios at 1550 nm. The design explores waveguide path length imbalance to realize ON/OFF switching behavior. The proposed MZI demonstrates compact footprint and serves as a building block for applications such as optical modulation, filtering, and switching.**

Index Terms— **integrated photonics, interferometer, modulation, optical switching, waveguides**

I. INTRODUCTION

The Mach–Zehnder interferometer (MZI) is one of the most widely used structures in integrated photonics due to its versatility in optical modulation, switching, and filtering. Its operation is based on splitting an optical signal into two arms, introducing a relative phase shift via path length or refractive index variations, and recombining the beams to achieve constructive or destructive interference. This simple yet effective configuration enables precise control of light amplitude and phase, making MZIs a cornerstone for applications in high-speed optical communications [1], biosensing [2], and quantum information processing [3]. Furthermore, their compatibility with planar waveguide technologies supports compact and scalable integration for advanced photonic circuits.

II. THEORY

The operation of an interferometer relies on the principle of optical interference, where two coherent light beams recombine after traveling different optical path lengths. In integrated photonics, the Mach–Zehnder Interferometer (MZI) is a common configuration due to its straightforward implementation and clear spectral response [4]. When an input optical field is split into two arms with path lengths L1 and L2, the output intensity is governed by the relative phase difference between the two arms. This phase difference is given by

$$\Delta\phi = \frac{2\pi}{\lambda} n_{eff}\, \Delta L$$

where λ is the free-space wavelength, n_{eff} is the effective refractive index of the waveguide mode, and ΔL is the path length imbalance. The transmission spectrum of an MZI is therefore periodic in wavelength, with constructive and destructive interference determined by $\Delta\phi$.

The periodicity of the transmission spectrum, known as the free spectral range (FSR), is a key measurable quantity. The FSR is related to the group index of the waveguide, n_g, by

$$FSR = \frac{\lambda^2}{n_g \Delta L}$$

Here n_g is defined as

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$$

which accounts for material and waveguide dispersion [5]. By fabricating MZIs with a known imbalance, ΔL, and measuring the resulting FSR from the transmission spectrum, one can experimentally extract n_g. This approach is widely used to validate numerical simulations of photonic waveguides and to study how geometry (e.g., width, height) and polarization (TE or TM) influence dispersion [6].

For the present design, the default strip waveguide geometry (220 nm height, ~500 nm width) will be used, though variations in width and splitter type can be explored to examine trade-offs in bend loss, coupling efficiency, and fabrication tolerances. Ensuring that the interferometer's FSR is smaller than the experimental measurement bandwidth (~50 nm) is essential for accurately resolving the interference fringes and extracting the group index.

III. MODELING AND SIMULATION

1) Waveguide

Using Lumerical MODE, the waveguide geometry was defined with a height of 220 nm and a width of 500 nm (Figure 1), and an eigenmode solver was applied around this structure. These dimensions are representative of standard values commonly employed in industry and wafer foundries, as they provide a good balance between fabrication feasibility and optical confinement. For this geometry, a modal analysis simulation was carried out in Lumerical MODE at the telecom wavelength of 1550 nm (Figures 2 and 3). The simulation allowed the identification of supported modes and provided key parameters such as the effective and group indices. These values are summarized in Table 1.

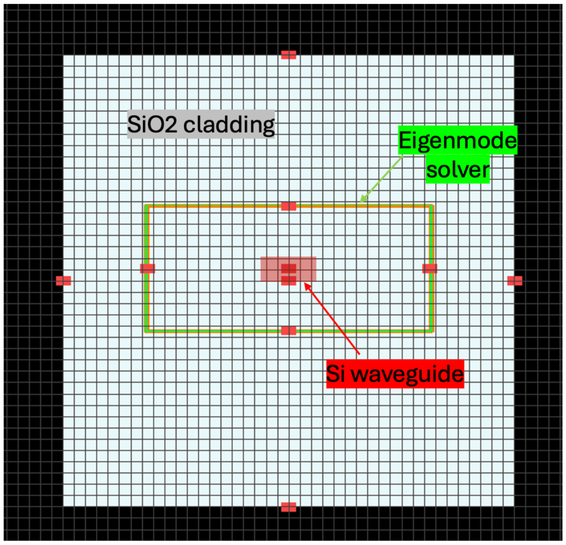


Fig. 1. The strip waveguide geometry (height=220 nm, width=500nm)

As shown in Figure 4, the imaginary component of the effective index is negligible and can be ignored (~10^{-9}). A wavelength sweep was then carried out from 1500 nm to 1600 nm, incorporating both material and waveguide dispersion. These two factors are essential considerations in the design of any photonic device that is produced.

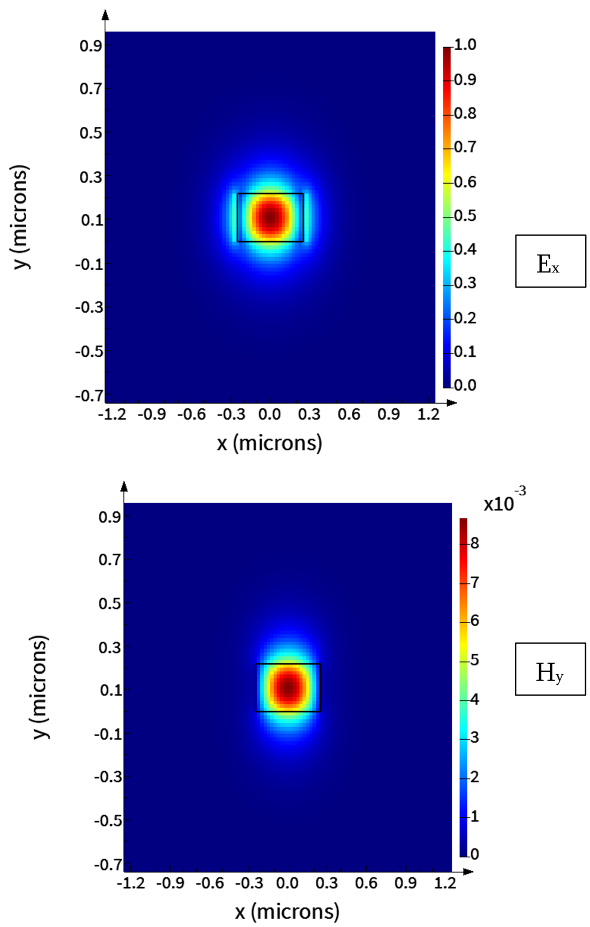


Fig. 2. Mode profiles (dominant components) of the fundamental TE mode.

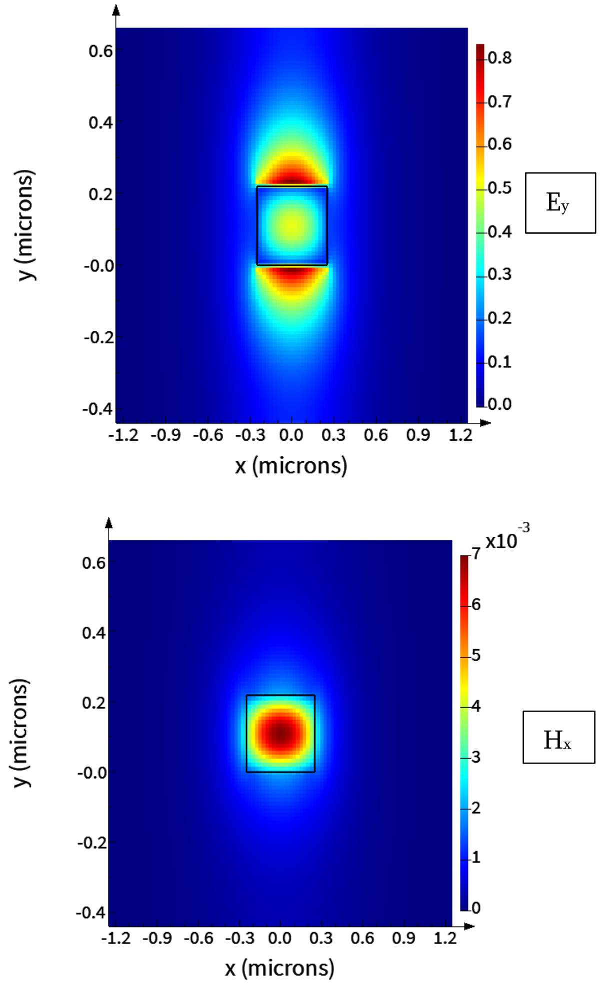


Fig. 3. Mode profiles (dominant components) of the fundamental TM mode.

The simulated strip waveguide with a cross-section of 220 nm height and 500 nm width has both TE and TM fundamental modes with effective indices of 2.502 and 1.847, respectively (Table 1). The higher effective index of the TE mode indicates stronger confinement of the optical field within silicon, consistent with the horizontal orientation of the electric field relative to the wider waveguide dimension. In contrast, the TM fundamental mode exhibits a lower effective index, reflecting weaker confinement and a greater fraction of the optical field leaking into the surrounding cladding. These differences in effective index highlight the waveguide's polarization dependence, which is crucial in device design, as it directly impacts mode propagation, dispersion, and coupling efficiency in integrated photonic circuits.

Tab. 1. Effective indices of TE and TM fundamental modes of the simulated Strip Waveguide (220 nm x 500 nm).

Mode	Effective Index
TE fundamental	2.502
TM fundamental	1.847

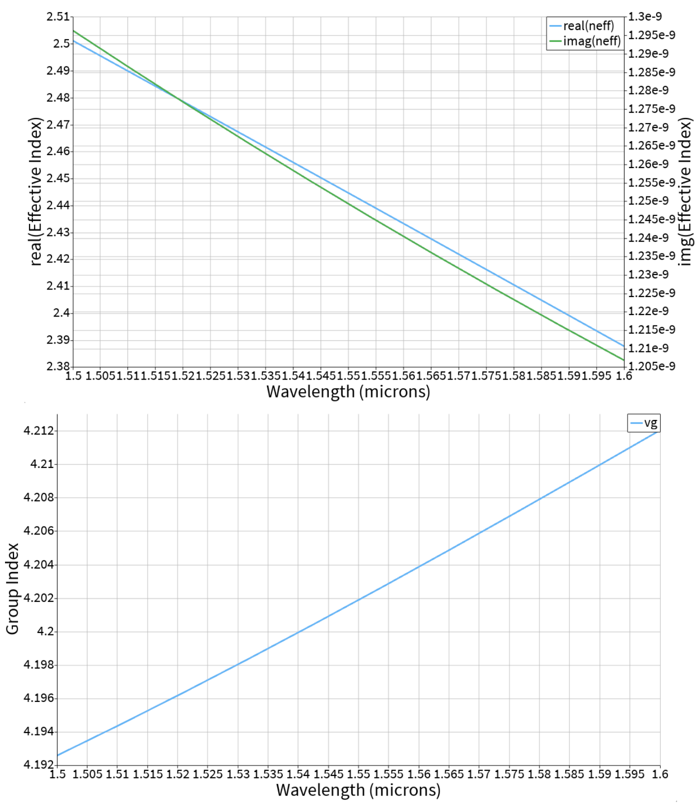


Fig. 4. Effective index and group index plots

Based on the modal analysis results above, the effective index was fitted as a function of wavelength. Using a truncated second-order Taylor expansion, the effective index can be expressed as:

$$n_{eff} = 2.44 - 1.13*(\lambda-1.55) -0.06*(\lambda -1.55)^2$$

where λ_0 is 1.55 um.

2) Mach-Zehnder Interferometer (MZI)

Transfer function of MZI:

$$T_{MZI} = \frac{1}{2}(1 + \cos(\beta\Delta L))$$

Tab. 2. Free spectral Range (FSR) data of corresponding path length differences of strip waveguide (220 nm x 500 nm) in MZI.

ΔL (μm)	FSR (nm)
200	2.85
100	5.75

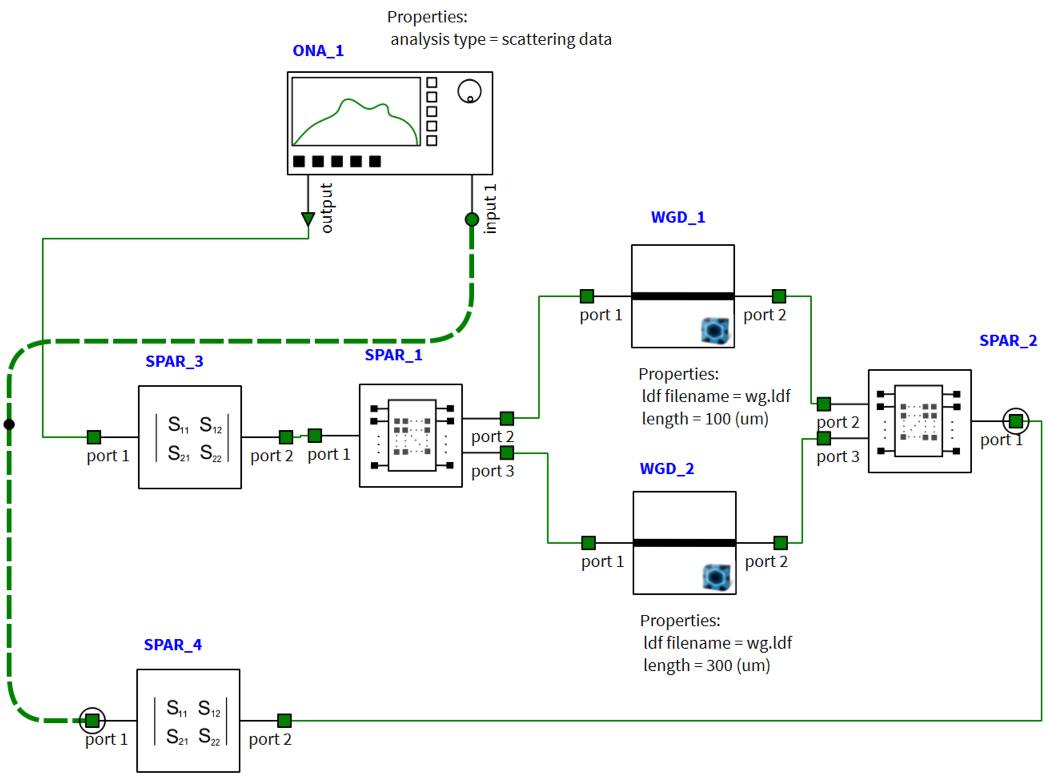


Fig. 5. MZI model implemented in the Lumerical INTERCONNECT software.

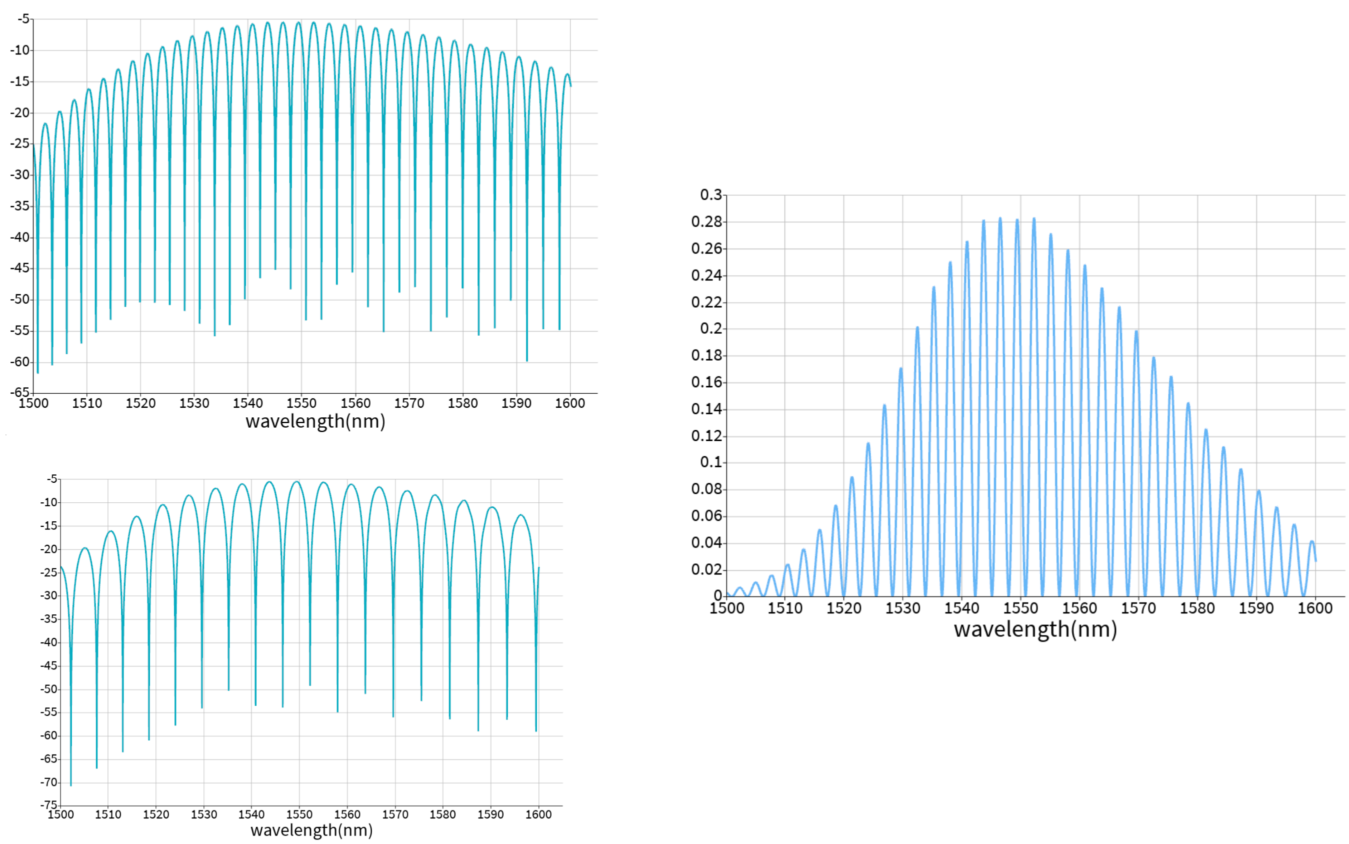


Fig. 6. TE gain of the MZI with cross-section of 500×220 nm2 for two different path lengths (ΔL= 200 μm and 100 μm), and for the fundamental TE mode

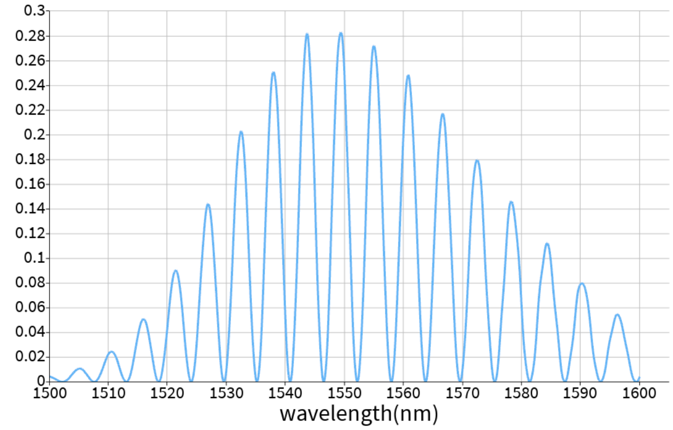


Fig. 7. Transmission spectrum of the MZI with cross-section of 500×220 nm2 for two different path lengths (ΔL= 200 μm and 100 μm), and for the fundamental TE mode

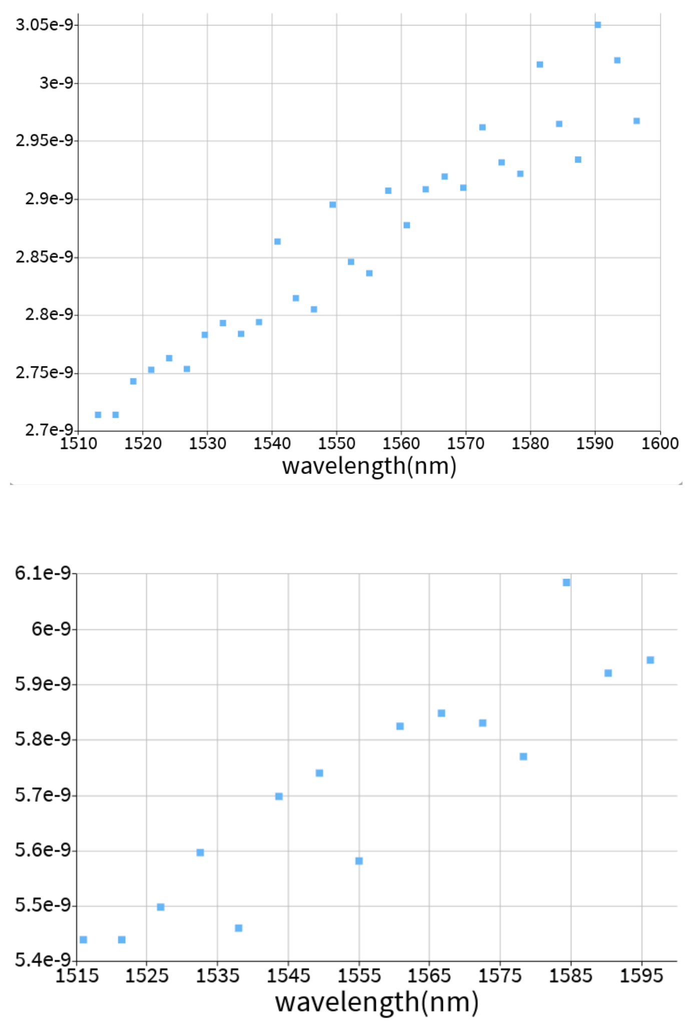


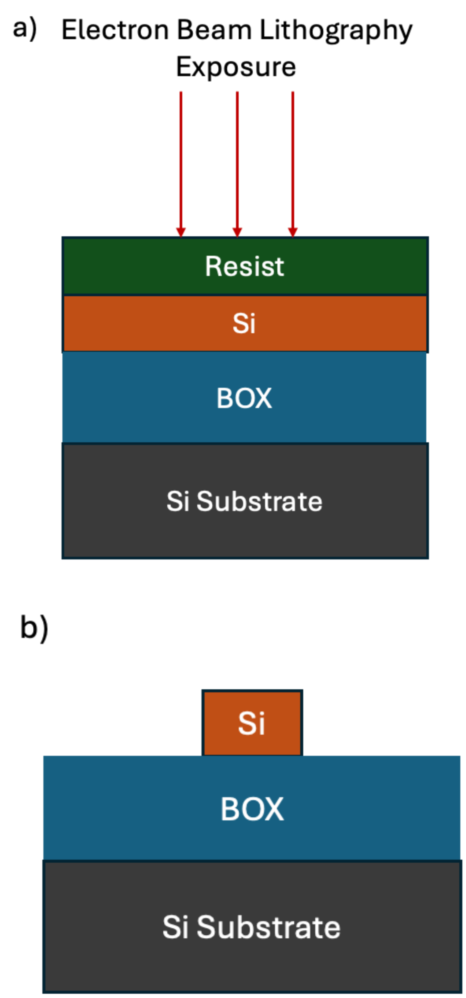
Fig. 8. The free spectral range (FSR) of the MZI with cross-section 500×220 nm^2 and ΔL= 200 μm (top) and 100 μm (bottom), respectively, for the fundamental TE.

IV. FABRICATION

1) *Fabrication Process*

The layout described earlier was sent for fabrication at Applied Nanotools Inc. (ANT), a facility specializing in prototype passive silicon photonic devices produced using Electron Beam Lithography (EBL). The fabrication workflow can be summarized as follows:

i) Selection of an appropriate SOI wafer;
ii) Deposition of a negative photoresist layer through spin coating followed by baking;
iii) Exposure of the wafer using EBL;
iv) Development of the resist by immersing the wafer in a hydroxide solution (TMAH) for a few minutes;
v) Rinsing and drying of the wafer;
vi) Removal of unexposed regions via plasma etching;
vii) Deposition of an oxide layer across the entire chip;
viii) Final dicing of the wafer into individual chips.



c)

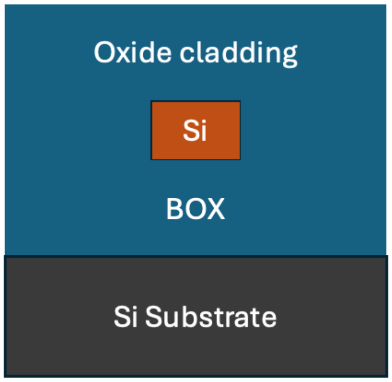


Fig. 9. Summary of the fabrication process. a) Wafer exposure (with a negative photo-resistive layer) for the electron beams (step iii); b) Cross-section after plasma etching (step vi); c) Oxide deposition (step vii).

2) *Manufacturing Challenges*

So far, we have discussed simulating process variations by considering the worst cases, namely the corners. But in reality, each of the process parameters has some statistical probability distribution. In the simplest model, the thickness of the wafer may be considered a Gaussian random variable with a mean and a standard deviation using data specified by the wafer manufacturer. If we know these distributions or can guess them, we can get a more realistic prediction of the circuit behavior using the Monte Carlo technique.

V. EXPERIMENTAL DATA

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

VI. CONCLUSION

This study presented the design and analysis of MZIs fabricated in SOI technology, using strip waveguides with dimensions of 220 nm in height and 500 nm in width.

All devices were produced via electron-beam lithography at the Applied Nanotools Inc. (ANT). Simulations for TE polarizations were carried out using Lumerical and MATLAB, while KLayout was employed for layout generation.

Key performance indicators included the group index and the free spectral range, which were used to evaluate the agreement between simulated and fabricated devices. The TE-mode waveguides demonstrated extremely accurate group index predictions, with errors below 0.01%.

Of the three devices analyzed, all exhibited group indices within the predicted range defined by corner analysis.

Acknowledgments

I/We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Omid Esmaeeli performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

Lastly, I gratefully acknowledge the financial support of PsiQuantum Inc. for enabling me to take this course.

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